Selecting a Reasoning Strategy: A Comparison Table

Introduction

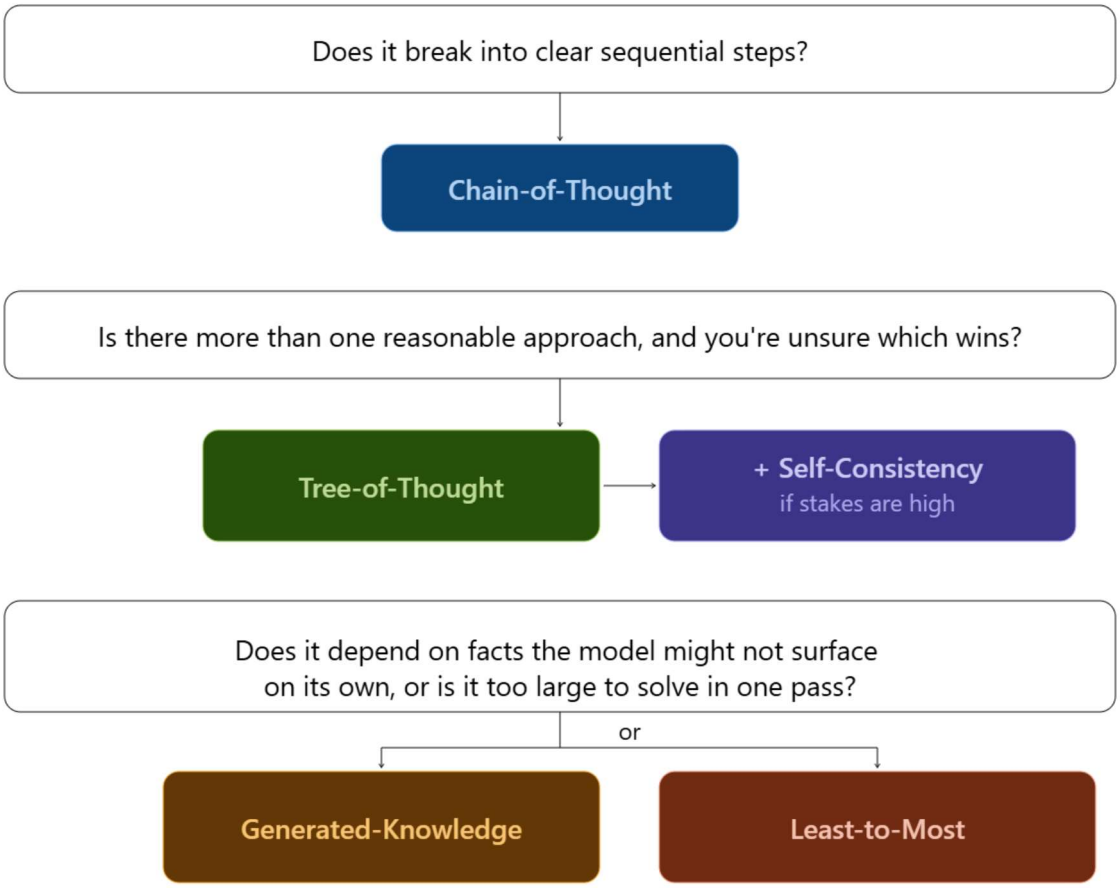
Five reasoning techniques exist, and real tasks often call for combining more than one rather than relying on a single approach. This reading is a single reference table: given a task in front of you, which technique (or combination) actually fits.

The full comparison

Technique	Path structure	Best for	Cost vs. direct
Chain-of-Thought	Single, sequential	Clear multi-step math/logic tasks	Low — one extra reasoning pass
Tree-of-Thought	Multiple, parallel	Tasks with several plausible approaches	High — explores multiple branches
Self-Consistency	Multiple, independent runs	High-stakes tasks needing reliable answers	High — runs the prompt several times
Generated-Knowledge	Single, knowledge-first	Tasks depending on facts the model might miss	Low-medium — one extra generation step
Least-to-Most	Single, staged	Large problems that decompose cleanly	Medium — multiple sequential steps

Matching technique to task shape

The fastest way to choose isn't memorizing the table — it's asking three questions about the task itself.



When to combine techniques

Some tasks benefit from stacking more than one technique rather than relying on a single approach.

Combination	Why it works
Generated-Knowledge □ Chain-of-Thought	Surface the facts first, then reason through them step by step
Least-to-Most □ Chain-of-Thought	Break the problem down, then reason through each sub-problem
Chain-of-Thought □ Self-Consistency	Get step-by-step reasoning, then run it multiple times for reliability

Combining techniques compounds their cost, so this is worth doing only when a single technique genuinely isn't enough — not by default.

A closing rule of thumb

> Start with the cheapest technique that could plausibly work (usually Chain-of-Thought). Add complexity — more branches, more knowledge, more runs — only when the task's difficulty or stakes actually demand it.

Summary

